

Problem-Solution Fit

Date	17 September 2026
Team ID	SWTID-2026-2311
Project Name	AI StudyBuddy API
Maximum Marks	2 Marks

Problem-Solution Fit

CUSTOMER SEGMENT(S) - College students preparing for exams with material scattered across notes, PDFs, textbooks and articles - Persona: Rahul – a second-year engineering student balancing exams and internship prep with limited time	PROBLEM - Lengthy, scattered study material is hard to understand and organize in limited time - Manual creation of flashcards, quizzes and revision notes is repetitive and time-consuming - No personalized study plan and no centralized, secure place for AI-generated resources
EXISTING ALTERNATIVES - Manually rewriting notes and building flashcards/quizzes by hand - Generic study-tips content that isn't derived from the student's own material	SOLUTION - Secure, JWT-authenticated backend for uploading study material and storing it centrally - Google Gemini-powered summary, flashcard, quiz and study-plan generation from that same material
FIT VALIDATION — The solution directly answers the problem: a single authenticated Material record replaces scattered files, and the four Gemini-backed generation endpoints replace hours of manual summarizing, flashcard-writing, quiz-setting and schedule-planning with one upload and a few API calls.	